Sahaj Bhadja

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Summary

Cloud solutions engineer with 2+ years designing **AWS/Azure architectures, microservices, APIs, integrations, and real-time systems**; experienced gathering requirements, documenting technical designs, coordinating deployments, troubleshooting production issues, and presenting solutions to customers.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Assessed user needs and designed **scalable multi-tenant cloud architecture** across Azure SQL, PostgreSQL, SQL Server, FastAPI, and storage services, supporting 500+ users while reducing dashboard latency 71%.
- Documented requirements, data flows, validation controls, and deployment processes for 5,000+ monthly integrations, sustaining 99%+ reliable production operations through reconciliation, failure isolation, monitoring, and CI/CD-backed releases.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Translated stakeholder workflows into **React interfaces, Flask services, and REST integrations** for uploads, processing, reporting, and feedback, serving 150+ users and 10,000+ daily requests on AWS.
- Integrated OpenAI, Claude, and Qwen behind adaptable APIs, directly validating architecture and application behavior while coordinating requirements, testing, production deployment, issue resolution, and documentation with technical stakeholders.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Designed **cross-platform microservices and third-party integrations** across AWS, Windows, Linux, and macOS, improving response times 30%, pipeline efficiency 68%, and data consistency through APIs, ETL, and caching.
- Partnered with Microsoft and GitHub engineers to gather requirements, review detailed designs, troubleshoot integration failures, and automate builds, installations, unit tests, integration tests, deployments, and operational reports.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Architected **secure real-time communication and information services** across React, Node.js, REST APIs, WebSockets, eight microservices, and Azure SQL, securely synchronizing 2M+ records with authentication and role-based access.
- Coordinated global Docker and Jenkins deployments while troubleshooting with Sentry, Azure Monitor, and Kibana, achieving 95%+ build validation, reducing deployment time 50%, and supporting reliable production releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Worked directly with customers to analyze technical requirements and present **end-to-end system designs** covering React interfaces, Flask APIs, Firebase synchronization, external integrations, authentication, payments, and operational reporting.
- Mapped and documented 12+ complex integration flows, implemented secure webhooks and real-time notifications, and supported deployment and service issues across 50,000+ records with sub-200-millisecond API responses.

Skills

Architecture: Requirements gathering, business analysis, system design, technical specifications, architecture diagrams
Cloud & Infrastructure: AWS, Microsoft Azure, IaaS/PaaS, microservices, Docker, Kubernetes, Linux, Windows
Integration: REST APIs, WebSockets, webhooks, real-time notifications, TCP/IP, authentication, access control
Technology: Python, Java, JavaScript, TypeScript, Node.js, SQL Server, Azure SQL, PostgreSQL, Firebase
Delivery: Customer presentations, deployment planning, troubleshooting, monitoring, Git, Jenkins, CI/CD, Agile

Projects

HMIS Live | *Cloud Architecture, APIs, Monitoring*

- Designed a **mission-oriented healthcare information system** with consent-based access, provider dashboards, appointments, REST integrations, real-time notifications, and Sentry monitoring, halving record retrieval time while reducing administrative workload 30%.

IoT Authentication Analysis | *IoT, Security, System Design*

- Designed and validated **secure IoT device authentication** across three user scenarios and four permission tiers, documenting integration assumptions and replacing centralized verification with tamper-resistant identity workflows while reducing repeated key checks 50%.

Chat2SQL Code | *Integration, Docker, Cross-Platform*

- Enhanced a **cross-platform integration application** supporting SQL Server and 10+ data systems with schema exploration, compatibility checks, query generation, Docker configurations, and reusable development and deployment workflows.